Akhil Theerthala

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Professional Experience

Senior Member Data Scientist | *Perfios Software Solutions* Apr 2025 – Present

- Leading uncertainty-quantification research for financial LLMs by building a BFSI-specific benchmark, auditing cultural and demographic biases, measuring calibration across numerical-precision conditions, probing cues associated with miscalibration, and testing steering, context conditioning, and fine-tuning as mitigation methods.
- Developing guardrails for LLM and agent applications in banking by mapping vulnerability surfaces, constructing adversarial and failure-mode evaluations, and testing output controls aligned with RBI's FREE-AI recommendations for safe, fair, explainable, and robust AI.
- Developed a market-intelligence support system for banking products, analyzing customer needs and competitive positioning to identify product white-space opportunities.
- Developed and now monitor a reference-free document image-quality assessment service using ViT regression; the production system identifies low-quality documents at **92% precision** before they enter HDB's downstream applications.
- Adapted **PaliGemma2** via LoRA on financial documents, achieving a **0.85 TEDS score** on internal benchmarks for structured document understanding.
- Received the **Circle of Excellence** award for research and contributions to generative AI and financial document intelligence.

Member Data Scientist | *Perfios Software Solutions* Jun 2023 – Apr 2025

- Distilled and quantized a multimodal document classifier into a lightweight student model, reducing latency from **8s** to **200ms** while preserving F1 parity in production-scale evaluation.
- Improved generalized table detection accuracy by **27.6%**, lifting downstream TSR module performance, via semi-synthetic data curation and systematic evaluation of YOLOv8 variants.
- Integrated a semantic row-detection module into the production TSR pipeline with **<40 ms** added turnaround time.

Publications

G. Matlin et al. (2026). **Data Attribution Should Guide Pretraining**
ATTRIB, NeurIPS 2026.

G. Matlin et al. (2026). **Role Steering of Language Models for Social Simulations**
Social Simulations Workshop, COLM 2026.

G. Matlin et al. (2026). **FinForge: A Semi-Synthetic Benchmark Generation Framework for Finance.**
AAAI 2026 Workshop on Agentic AI in Financial Services.

A. Theerthala (2025). **A Data-Centric Framework for Training Behaviour-Aware Personal Finance LMs.**
FinNLP, EMNLP 2025.

Selected Projects & Open-Source Work

Reasoning Dataset Creation Challenge Winner – 1st Place (Global)

- Won 1st place globally out of 150+ teams in the Reasoning Dataset Creation Challenge hosted by Bespoke Labs, Hugging Face, and Together.ai.
- Built synthetic data pipelines using real-world personal finance queries to create instruction-tuning and reasoning datasets tailored to Indian financial contexts.
- Fine-tuned 8B/14B models and released datasets, LoRA adapters, and quantized weights on Hugging Face, reaching 56k+ downloads across direct and community GGUF adaptations.

Density vs. Diversity – Data Curation Strategy Validation for VLMs Hugging Face Article

- Investigated trade-offs between dense and diverse sampling strategies by curating 15k-sample synthetic datasets for Vision-Language Models (VLMs).
- Conducted comparative analysis of 4B and 8B models across in-domain and real world benchmarks to assess reasoning and OOD generalization.

Moral Reasoning Dataset (Morality-as-Cooperation) GitHub

- Designed a five-stage LLM pipeline (fact extraction, stakeholder and impact analysis, framework reasoning, consolidation, response generation) that produces structured chain-of-thought traces for resolving moral dilemmas.
- Grounded the reasoning in Morality-as-Cooperation (MAC) theory, mapping each dilemma to seven cooperative domains (kinship, group loyalty, reciprocity, heroism, deference, fairness, property) for a cross-culturally supported structure instead of ad hoc ethical labels.

Causal Analysis of Social Media Signals on Crowdfunding Success GitHub

- Designed a multi-stage experimental framework to decouple the impact of social engagement metrics from fundamental content features on funding rates.
- Conducted stepwise ablation studies to quantify the marginal lift of social signals, demonstrating that content features drive success independently of engagement metrics

Additional Experience

Research Volunteer | *FSIL & HCAI Labs, Georgia Tech* Aug 2025 – Present
(External Collaborator)

- Co-authored **FinForge**, a semi-synthetic benchmark generation pipeline for financial tasks — accepted at the **AAAI 2026 Workshop on Agentic AI in Financial Services**.
- Contributed to **Role Steering of Language Models for Social Simulations**, published in the **Social Simulations Workshop at CoLM 2026** and **Data Attribution Should Guide Pretraining**, accepted into ATTRIB workshop, NeurIPS'26.
- Contributing to **Capabilibarav2**, an HCAI Lab training-data attribution project that traces which source documents influence large language model outputs.

Education

MS, Data Science Apr 2026 – Expected 2028
International Institute of Information Technology, Hyderabad

B. Tech, Aerospace Engineering Aug 2019 – May 2023
Indian Institute of Technology, Kharagpur

Skills

Core ML: Python, SQL, PyTorch, Scikit-learn, NumPy, Pandas
LLMs & Fine-Tuning: Hugging Face Transformers, TRL, PEFT/LoRA, vLLM, instruction tuning, structured generation, RAG
Agentic Systems & Evaluation: Smolagents, Google Agent Development Kit, LangChain, LangGraph, tool-use agents, multi-agent workflows, inference-time scaling, task-specific evaluation, failure-mode analysis
Data-Centric ML: synthetic data generation, benchmark construction, dataset curation, ablation studies, model comparison, evaluation design and benchmarking
Multimodal & Document AI: vision-language models, document classification, table structure recognition, visual quality estimation, OCR-adjacent pipelines
Optimization & Engineering: distillation, quantization, async inference pipelines, Docker, Git, Linux

Professional Development

Technical AI Safety (*BlueDot Impact*)
The Reasoning Course (*Hugging Face*)
Machine Learning in Production (*Coursera*)
Agentic AI (*DeepLearning.AI*)
Generative AI Nanodegree (*Udacity*)
Deep Learning Specialization (*DeepLearning.AI*)